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Embry-Riddle Aeronautical University, yowt@my.erau.edu

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INITIAL ORBIT DETERMINATION OF UNCOOPERATIVE SATELLITES USING PARTICLE SWARM OPTIMIZTION FROM THE POINT OF VIEW OF A NOVEL OBSERVER ORBIT

By

Taylor M. Yow

A Thesis Submitted to the Faculty of Embry-Riddle Aeronautical University

In Partial Fulfillment of the Requirements for the Degree of

Master of Science in Aerospace Engineering

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Embry-Riddle Aeronautical University

Daytona Beach, Florida

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This Thesis was prepared under the direction of the candidate's Thesis Committee Chair, Dr. [Jill Smith], Department of Aerospace Engineering, and has been approved by the members of the Thesis Committee. It was submitted to the Office of the Senior Vice President for Academic Affairs and Provost, and was accepted in the partial fulfillment of the requirements for the Degree of Master of Science in Aerospace Engineering.

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To every little girl who has looked up to the stars and been inspired.

You can do it.

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ABSTRACT

Initial orbit determination is an incredibly valuable tool in the field of space situational awareness (SSA). As the number of launches increases every year, more objects are being put into space around the Earth and with political tensions mounting between certain space-accessing countries such as the United States of America (USA), Russia, and China, the number of satellites whose orbits are unknown to a political rival are increasing as well. In this research, orbit estimation of uncooperative target satellites from the point of view of novel orbits using particle swarm optimization (PSO) algorithm is explored. Two cost functions for the PSO algorithm are investigated. The first cost function investigated uses the solution to Lambert's Problem to estimate the initial velocity of the target. The second cost function uses a full estimated state instead of just the position as used in the first cost function. Having the observer satellite in a novel orbit called Rock-Around orbit (RAO) about an area of interest expands the viewing opportunities for that area, allowing for more measurements of the target satellite and more confidence in the estimated orbit of the target. RAOs are based off flower constellation theory which allows for an increased observation area. The RAO of the observer allows the observer to oscillate about a region of interest, in which the target orbit resides. This research shows that the PSO algorithm with both cost functions can successfully estimate the initial state of the target.

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NOMENCLATURE

$\hat{L}$	Line of Sight Vector
μ	Gravitiation Parameter
ω	Argument of Perigee
Ω	RAAN
ρ	Range
θ	True Anomoly
a	Semi-Major Axis
e	Eccentricity
F	Force
G	Gravity Coefficient
m	Mass
r	Radius
t	Time

1 Introduction

With the number of launches from both commercial and government space agencies increasing every year, more objects are put into orbit. With political adversaries, such as the USA and Russia or China, launching reconnaissance satellites whose orbits are kept classified, the amount of unknown satellite orbits from each country's point of view are increasing. These means that determining an orbit of an uncooperative satellite is now a necessary step in ensuring space domain safety.

1.1 Problem Statement

On board space situational awareness (SSA) is going to become an integral part of new satellites as the space domain becomes more crowded. This research aims to prove that a particle swarm optimizer can provide and initial estimate of a target's orbit using measurements from an observer in a Rock-Around orbit (RAO). A satellite with a RAO provides more observation opportunities than a satellite not in an RAO. The RAO allows the observer to oscillate over, under, and to either side of a region of interest.

1.2 Motivation

The launch and operation of reconnaissance satellites is seeing a resurgence since the cold war, making the number of uncooperative objects in orbit higher than it has ever been.

Rank	Country	# of Military Satellites
1	United States of America	239
2	China	140
3	Russia	105
4	France	18
5	Italy	13
6	Israel	11
7	India	9
8	Germany	7
9	United Kingdom	6
10	Spain	6

Table 1.1 Countries that have the most military satellites ranked [3] [4].

Innovations in computer technology have lead to faster and more robust onboard computers capable of running orbit determination software autonomously on orbit. It is these conditions that have motivated the development of the method outlined in this thesis.

1.3 Outline of Thesis

Chapter 2 gives a summary of astrodynamics and and the background information necessary for orbit determination. Classical and modern methods of orbit determination are outlined each with an algorithmic pseudo-codes for the key methods. Also in Chapter 2 is a general summary of Particle Swarm Optimization (PSO). Chapter 3 outlines the theoretical method used in this research, with a focus on Rock Around Orbits and PSO methods used to estimate an initial state of a target. Chapter 4 provides the results of the two and six variable PSO algorithms outlines in Chapter 3 with varying amounts of noise. Chapter 5 compares the results of the two and six variable PSO for each noise level and discusses the error of each method. Possible additions and future work are also discussed in Chapter 5.

2 Background

This chapter includes four main topics of literature review. First, a review of fundamental astrodynamics. Second, an overview of different measurement types. Third, a synopsis of several methods of angles-only orbit determination. Lastly, a introduction of particle swarm optimization.

2.1 Astrodynamics Fundamentals

The term astrodynamics encompasses several different related topics. Celestial mechanics, the study of the dynamic motion of celestial bodies. Orbital dynamics which studies the motion of all orbiting bodies. Attitude dynamics examines the orientation of an object in space. The following definition of astrodynamics from *Space Vehicle Design* [5] combines elements of celestial mechanics, orbital dynamics, and attitude dynamics.

Astrodynamics is the study of the motion of man-made objects in space, subject to both natural and artificially induced forces.

Since astrodynamics is the study of man-made objects in space, the study of astrodynamics has only existed since the 1950's. Therefore its predecessors (astronomy, celestial mechanics, and orbital dynamics) must be examined along side of it.

2.1.1 History

Many civilizations around the world looked to the heavens long before the big names of astronomy came onto the scene.

Claudius Ptolemaeus (100-170 A.D.), or Ptolemy as he is more commonly known, published *Almagest* in 100 A.D. Within *Almagest* was thirteen individual books which described his theory of an Earth-centered solar system. Ptolemy's excentric and epicycle systems of planetary motion were extremely complex so he included tables in the *Almagest* with predictions of the planets positions from his models. These tables are known as the *Ptolemaic Tables*. After Ptolemy published his theories there were no major changes to the field of astrodynamics for almost a millennium.

Nicholas Copernicus (1473-1543) is most famously known for his heliocentric, or sun-centered, model of the solar system published in *De Revolutionibus Orbium Coelestium [On the Revolutions of the Celestial Sphere (1543)].* The Copernican system alleviated many of the difficulties with the Ptolemaic system, but since Copernicus assumed that planetary orbits were circular about a central point the problem was not entirely solved .

Galileo Galilei (1564-1642) built upon Ptolemy and Copernicus' works with the use of the telescope. In his *Dialog on the Two Chief Systems of the World, the Ptolemaic and Copernican* (1632) Galileo used his telescopic observations to defend the validity of the Copernican system [6].

The Danish nobleman Tycho Brahe (1546-1601) is widely known for his unprecedented accurate astronomical observations. In 1573 he published his observations on a new star, *De Nova Sella*, contradicting the Aristotelian cosmological belief that the heavens were unchanging [7]. In 1576 Tycho acquired funding from the Danish King Frederick II to construct two observatories on the island of Hven off the coast of Copenhagen. The observatories Uraniborg and Stjerneborg gave Tycho the ability to make simultaneous, independent observations possible. Tycho became the Imperial Mathematician to the court of Emperor Rudolph II in Prague in 1599. In this position he hired Johannes Kepler (1571-1630) to help him work out a mathematical proof of his Tychonic system, which was slightly different than the Ptolemaic and Copernican systems, using his observations of Mars. Unfortunately Tycho died before he could finish his work and left all his astronomical observations to Kepler, who succeeded him as Imperial Mathematician.

Kepler published *Astronomia Nova (New Astronomy)* in 1609 using Tycho's data on Mars. Included were his first two laws. In 1619 Kepler published his third law in *Harmonices Mundi Libri V (Harmony of the World)* completing his description of orbital motion. In the preface Kepler wrote "So I throw the dice and write a book for my contemporaries or for postarity. I do not care. It may take a hundred years for my book to find readers, it even took God 6000 years to find and observer." [8]

Kepler's Laws

Law I: The orbit of each planet is an ellipse with the Sun at one focus.

Law II: The line joining the planet to the Sun sweeps out equal areas in equal times.

Law III: The square of the period of a planet is proportional to the cube of its mean distance to the Sun.

Kepler's Laws fully described the kinematics of orbital motion, but it was Issac Newton (1642-1727) that developed the laws that described the dynamics. While Cambridge was shut down due to the bubonic plague, Newton stayed busy with several important problems [9]. His famous falling apple observation prompted the thought of what force would cause this motion and how far that force extended, particularly to the Moon. Unfortunately, Newton's calculations for the Moon were about 15% off. He quickly abandoned the problem and focused on developing the binomial theorem, early calculus, and the foundations of the study of white light. After graduating with his Ph.D. from Cambridge, Newton was encouraged by Edmond Halley (1656-1742) to continue his research of planetary motion. Newton finished his work in 1686 and Halley paid for Newton's manuscript to be printed in 1687. Newton's *Philosophiae Naturalis Pricipia Mathematica (Mathmatical Principles of Natural Philosophy, or Principia)* introduced his three laws of motion, which completed the solution of orbital motion [6].

Newton's Laws

Law I: Every body continues in its state of rest, or of uniform motion in a straight line, unless it is compelled to change that state by forces impressed upon it.

Law I: The change of motion is proportional to the motive force impresses and is made in the direction of the right line in which that force is impressed.

Law I: To every action there is always opposed and equal reaction: or, the mutual actions of two bodies upon each other are always equal and directed to contrary parts. [10]

The fundamental equations of orbital mechanics come from the application of Newton's second law and Newton's law of universal gravitation, which states that the force between two particles is directly proportional to the product of their masses and inversely proportional to the square of the distance between the particles (Equation 2.1).

$$\boldsymbol{F} = -G\frac{m_1 m_2}{r^2}\hat{\boldsymbol{r}} \tag{2.1}$$

To derive the mathematical model of two-body orbital motion, or unperturbed Keplerian motion, Newton's second law and the Law of Universal Gravitation are directly applied.

2.1.2 Two-Body Problem

The Two-Body Problem refers to the mathematics of orbital motion that follows unperturbed Keplerian motion. Kepler's first and second laws describe the physical conditions for the two body motion, elliptical motion around a central body and equal areas sweep out in equal times. The setup of two-body motion is illustrated in Figure 2.1 There are four major assumptions that must be made for the application of Newton's Laws to the two-body dynamical system.

Assumptions for the Two-Body Problem

1. The mass of the satellite is negligible compared to the mass of the body.
2. The default coordinate system is inertial.
3. Both the satellite and the body are spherically symmetric with uniform density.
4. The only forces that act on the system are gravitational forces acting along the line joining the centers of the two bodies.

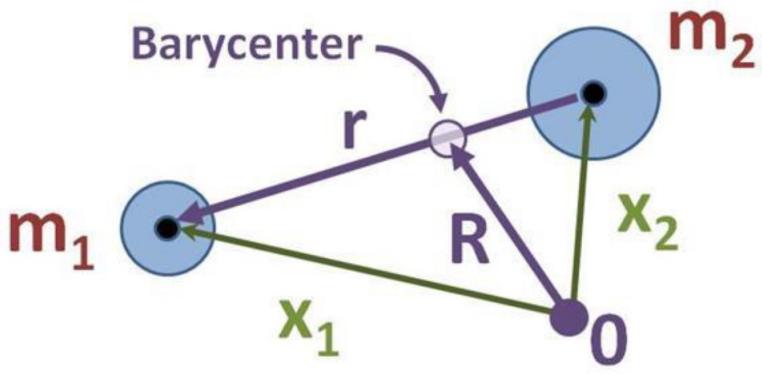


Figure 2.1 Representation of the Two-Body Problem setup.

The first assumption stems from the fact that the mass of any orbiting satellite will be significantly less than the mass of the body it is orbiting, this allows for the gravitational force acting on the main body from the satellite can be ignored. The second assumption, that the inertial coordinate system is used, allows for regular vector calculus operations. The third assumption allows the application of Newton's laws to the system by modeling the two bodies as point masses. The final assumption allows for all other forces besides the gravitational force from the larger body to be considered small enough in magnitude or slow enough over the considered time, which allows them to be ignored.

Keeping these assumptions in consideration, the vector equation for describing a satellite in the two-body problem can be derived by applying Newton's second law to the bodies and using the Universal Law of Gravitation as the force term. With this method the fundamental equation for unperturbed two-body motion, Equation 2.2, is obtained.

$$\ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\boldsymbol{r} \tag{2.2}$$

This is the mathematical model used for satellite motion utilized by all of the classical orbit determination methods.

2.2 Angles Only Orbit Determination

The process of calculating an initial orbit estimation using only angular measurements is known are angles-only orbit determination (AIOD). In the context of this research, optical measurements (i.e images taken from an on-board camera via the observer satellite) are assumed to be the source of angular observation. Since range data is not included in these optical measurements, additional observations and/or special processing must be used in this type of orbit determination compared to other methods of orbit determination. From each image two angular measurements can be obtained, either azimuth and elevation angles or spherical coordinate angles (i.e. right ascension and declination). Since each image provides a maximum of two angles, a minimum of three unique images with two angular measurements each must be obtained to determine a satellite's state.

2.2.1 AIOD Method Requirements

A general set of requirements for angles-only orbit determination exists despite the numerous methods of execution. These requirements are based off the observation geometry of the problem. Assuming that the location of the observation site, $\mathbf{R_i}$,and the line of sight (LOS) vector, $\mathbf{L_i}$, to the observed target are known, an expression for the position vector to the target, $\mathbf{r_i}$, can be derived. Figure 2.2 demonstrates the geometry of a typical AIOD setup.

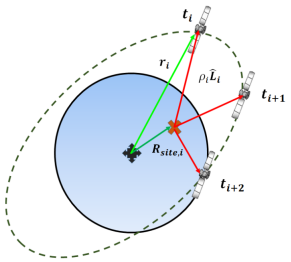


Figure 2.2 Representation of the vector setup for AIOD Process [1].

To mathematically describe the geometry of each observation, Equation (2.3) is used. Where the ith subscript denotes the ith observation, for $i = 1, 2, ..., N$, and N denotes the

number of total observations.

$$\mathbf{r_i} = \rho_i \hat{\mathbf{L}}_\mathbf{i} + \mathbf{R_i} \tag{2.3}$$

To describe satellite motion in 3D space, a state vector with six independent quantities must be determined. The six quantities of the state vector are most commonly in the form of position and velocity vectors or the classical Keplerian orbital elements. Figure 2.3 illustrates each of the orbital elements and Table 2.1 gives a short definition of each.

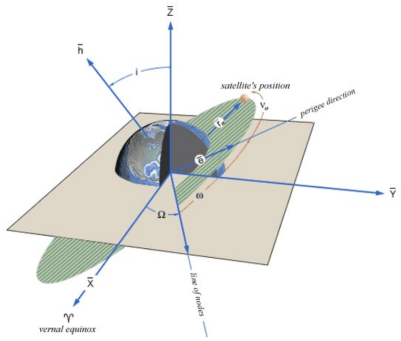


Figure 2.3 Illustration of classic Keplerain orbital elements [1].

Table 2.1 Classical Keplerian Orbital Elements

Orbital Element	Symbol	Definition
Semi-major axis	a	Size of the orbit.
Eccentricity	e	Shape of the orbit.
Inclination	i	Angle between the Earth's axis of rotation and the orbit's angular momentum vector.
Argument of Perigee	ω	Angle defining the closest approach point (perigee) of the orbit with respect to the surface of the Earth.
Right Ascension of the Ascending Node	Ω	Angle between the intersection of the orbit and equatorial planes.
True Anomaly	v, θ,f	Angle between the satellite's location in the orbit and the perigee of the orbit.

The AIOD process is used to determine a satellite's state from angular observations. A singular image provides a maximum of two independent angles $\{\alpha_i, \delta_i\}$ that are used to form

the Line of Sight (LOS) vector. To determine the six state variables of satellite motion, a minimum of three independent optical measurements are needed. Most historical methods of orbit determination use the minimum number of observations.

Once the observations are acquired, the unknown range value ρ_i that corresponds to each observed LOS vector must be determined. Different methods work best with different spacings between measurements. For example, Gauss' method works best when measurements are separated $\leq 60°$. Escobal's method can have widely spaced measurements though, even accounting multiple revolutions of a satellite between measurements.

2.3 Historical Methods

This section discusses some of the most important orbit determination methods throughout history. Two of the earliest methods for solving the AIOD problem came from Laplace and Gauss. Originally used for determining the motion of solar system bodies such as the planets, these methods were not intended to be used with close orbiting bodies such as the artificial satellites of today.

2.3.1 Laplace's Method

A French mathematician and astronomer, Pierre-Simon de Laplace (1749-1827), developed one of the first successful AIOD methods. Laplace's method works by estimating the state (positon and velocity) of the target at the middle observation time. The method requires the minimum of three observations with two angular measurements each to determine the six state variables of the target.

Laplace's Method Pseudo-code

---

1. 1: Given values: $\hat{\mathbf{L}}_\mathbf{i}$, $\mathbf{R_i}$ for i = 1,2,3.

2. Form primary vector equation for orbit geometry:

$$\mathbf{r_i} = \rho_i \hat{\mathbf{L}}_\mathbf{i} + \mathbf{R_i} \tag{2.4}$$

3. Double differentiating Equation (2.4) and rearranging terms produces Equation (2.5).

$$\ddot{\rho}\hat{\mathbf{L_i}} + 2\dot{\rho}\dot{\hat{\mathbf{L}}}_\mathbf{i} + \rho(\ddot{\hat{\mathbf{L}}}_\mathbf{i} + \frac{\mu}{r^3}\hat{\mathbf{L}}_\mathbf{i}) = -\ddot{\mathbf{R}}_\mathbf{i} - \frac{\mu}{r^3}\hat{\mathbf{R}}_\mathbf{i} \tag{2.5}$$

4. Since all derivative terms are unknown, an approximation for the derivatives of $\mathbf{r}$ and $\hat{\mathbf{L}_2}$ can be determined using Lagrange's interpolation formula with the assumption that the middle time is zero.

5. Assuming that observations are from a singular site, determine the derivatives of the site vector:

$$\dot{\mathbf{R}}_\mathbf{2} = \omega_\mathbf{Earth} \times \mathbf{R_2} \tag{2.6}$$

$$\ddot{\mathbf{R}}_\mathbf{2} = \omega_\mathbf{Earth} \times \dot{\mathbf{R}}_\mathbf{2} \tag{2.7}$$

6. If observations are from multiple sites, use the Herrirck-Gibbs method outlined in section 6.6.2 of Vallado [6] .

7. After 5, we have a matrix vector equation with unknowns ρ,$\dot{\rho}$,$\ddot{\rho}$, and r.

$$\begin{bmatrix} \hat{\mathbf{L}}_\mathbf{i} & 2\dot{\hat{\mathbf{L}}}_\mathbf{i} & \ddot{\hat{\mathbf{L}}}_\mathbf{i} + \frac{\mu}{r^3}\hat{\mathbf{L}}_\mathbf{i} \end{bmatrix} \begin{bmatrix} \ddot{\rho} \\ \dot{\rho} \\ \rho \end{bmatrix} = - \begin{bmatrix} \ddot{\mathbf{R}}_\mathbf{i} + \frac{\mu}{r^3}\mathbf{R_i} \end{bmatrix} \tag{2.8}$$

8. By applying Cramer's rule, an eighth order polynomial with unknown vector magnitude r can be formed. Before initializing the iterative process, a guess for r must be made. The real root gives the desired correct root to the equation and leads to the determination of the position vector for middle time [11].

9. Determine the velocity at middle time:

$$\mathbf{v_2} = \dot{\rho}_2\hat{\mathbf{L}_2} + \rho\dot{\hat{\mathbf{L}}}_\mathbf{2} + \dot{R}_2 \tag{2.9}$$

10. The full state vector $(\mathbf{r_2}, \mathbf{v_2})$ at the middle observation time has now been determined.

The accuracy of Laplace's method for Earth orbiting satellites is unreliable and therefore is primarily used for distant planetary or comet observations.

2.3.2 Gaussian Determination

Lagrange-Gibbs f and g Solutions

To fully understand Gauss's orbit determination algorithm, the Lagrange-Gibbs f and g functions must also be understood. The Lagrange-Gibbs f and g functions are scalar functions that allow the position and velocity of a satellite at any time to be written as a linear combination of the position and velocity at a specific time. The f and g solution starts with a couple notes/assumptions.

1. The f and g functions are approximations of an infinite series.
2. Assume orbital motion is purely planar motion.

A linear combination of the position and velocity at an initial time t_0 using the scalar functions f and g and their derivatives gives the position and velocity at any time along an orbit (Equations (2.10) and (2.11)).

$$\mathbf{r}(t) = f\mathbf{r}(t_0) + g\mathbf{v}(t_0) \tag{2.10}$$

$$\mathbf{v}(t) = \dot{f}\mathbf{r}(t_0) + \dot{g}\mathbf{v}(t_0) \tag{2.11}$$

In order to use the f and g functions (Equations(2.12) - (2.15)) , several orbital quantities also need to be calculated. These quantities include the semi-major axis (a), magnitude of position vector (r), and eccentric anomaly (E). Where $\hat{E} = E - E_0$is defined as the change in eccentric anomaly between the desired and initial epoch times, t and t_0 respectively.

$$f = 1 - \frac{a}{r_0}(1 - cos(\hat{E})) \tag{2.12}$$

$$\dot{f} = \frac{\sqrt{\mu a}}{r r_0} sin(\hat{E}) \tag{2.13}$$

$$g = (t - t_0) + \frac{a^{3/2}}{\mu}(sin(\hat{E}) - \hat{E}) \tag{2.14}$$

$$\dot{g} = 1 - \frac{a}{r}(1 - cos(\hat{E}) \tag{2.15}$$

With sufficient background on the f and g functions, Gauss' orbit determination technique can be discussed. The following algorithm summarizes Gauss' method.

Gauss' Method Pseudo-code

---

1. Givens $\hat{\mathbf{L}}_\mathbf{i}$, $\mathbf{R_i}$, for i = 1,2,3.

2. Assuming all position vectors are co-planer

$$c_1\mathbf{r_1} + c_2\mathbf{r_2} + c_3\mathbf{r_3} = 0 \tag{2.16}$$

3. Take cross products of Equation ((2.16)) to define expressions for c_1, c_2, c_3 with $c_2 \neq 0$.

4. Write the position vectors in terms of the desired time (t_2) using the f and g functions.

$$\mathbf{r_i} = f_i\mathbf{r_2} + g_i\mathbf{v_2}, i = 1, 3 \tag{2.17}$$

5. Re-scale the coefficients using (2.17) and letting $c_2 = -1$.

$$c_1 = \frac{g_3}{f_1 g_3 - f_3 g_1} \tag{2.18}$$

$$c_3 = \frac{-g_1}{f_1 g_3 - f_3 g_1} \tag{2.19}$$

6. Substitute in the series form of the f and g functions and rewrite c_1, c_2, c_3.

7. Substitute Equation (2.3) into Equation (2.16).

8. Now include the observed LOS vectors $\hat{L}_i$ to form a matrix vector equation similar to Equation (2.8) in Laplace's Method.

$$\begin{bmatrix} \hat{\mathbf{L_1}} & \hat{\mathbf{L_2}} & \hat{\mathbf{L_3}} \end{bmatrix} \begin{bmatrix} c_1\rho_1 \\ c_2\rho_2 \\ c_3\rho_3 \end{bmatrix} = \begin{bmatrix} \mathbf{R_1} & \mathbf{R_2} & \mathbf{R_3} \end{bmatrix} \begin{bmatrix} -c_1 \\ -c_2 \\ -c_3 \end{bmatrix} \tag{2.20}$$

9. Solving the equations simultaneously using matrix methods results in an eighth order polynomial similar to that obtained in Laplace's Method except more complicated by the presence of several positive, real roots. Each root much be inspected in order to determine the correct solution.

10. Iteration: Go back to step 5 and instead of using the series approximation of the f and g functions, use the actual f and g functions. Solve the matrix vector equation (2.20) again and repeat until the slant-range values converge.

Gauss' Method provides and accurate method for satellite orbit determination given that the measurements are ideally spaced apart. Extensions to this method allowing for multiple measurements exist.

2.3.3 Lambert's Problem

Lambert's Problem in its simplest form is finding the orbit between two positions when given the time of flight between them. Though it is not necessarily a AIOD process, Lambert's Problem is commonly used in orbit determination processes. A Swiss mathematician, Johann Heinrich Lambert (1728-1777), was especially fond of geometry, studying projection mapping, hyperbolic functions, and conic sections. Lambert proposed the following theorem stating that the time of flight between two points on an orbit depends entirely on the geom-

etry of the orbit itself.

Lambert's Problem Pseudo-code

---

1. Givens: $\mathbf{r_1}$, $\mathbf{r_2}$, t_1, t_1, with $t_1 > t_1$.

2. Define necessary quantities needed for later calculations.

   - Chord between position vectors:

$$c = \frac{\mathbf{r_2} - \mathbf{r_1}}{\|\mathbf{r_2} - \mathbf{r_1}\|} \tag{2.21}$$

   - Transfer angle between $\mathbf{r_1}$ and $\mathbf{r_2}$:

$$\theta = f_2 - f_1 \tag{2.22}$$

   - Semiperimeter (s) of the triangle formed by the two positions vectors and the focus.

$$s = \frac{r_1 + r_2 + c}{2} \tag{2.23}$$

   - Parameters α and β:

$$\sin\frac{\alpha}{2} = \sqrt{\frac{s}{2a}} \tag{2.24}$$

$$\sin\frac{\beta}{2} = \sqrt{\frac{s-c}{2a}} \tag{2.25}$$

3. Determine the shape of the orbit (elliptic/parabolic) by finding the parabolic transfer time t_p.

$$t_p = \frac{\sqrt{2}}{3\sqrt{\mu}}[s^{3/2} - sgn(\sin\theta)(s-c)^{3/2}] \tag{2.26}$$

4. If $t_2 - t_1 > t_p$, the transfer orbit is elliptic. If otherwise, the orbit is either parabolic

or hyperbolic.

5. Find the correct value of α via calculating the minimum energy transfer time, t_m.

$$t_m = \sqrt{\frac{s^3}{8\mu}}(\pi - \beta_m + \sin\beta_m \tag{2.27}$$

6. Using the value of θ, find the values of the parameters α and β.

7. Solve Lambert's Equation (2.28)using an iterative numerical approach.

$$t = \sqrt{\frac{a^3}{\mu}}(\alpha - \sin\alpha - (\beta - \sin\beta)) \tag{2.28}$$

8. Find the velocities at t_1 and t_2 using the unique solution for a.

9. The orbit is now fully defined by a full state, either $\{\mathbf{r_1}, \mathbf{v_1}\}$ or $\{\mathbf{r_2}, \mathbf{v_2}\}$.

Gooding's method uses Lambert's Problem to iterate on different guesses for unknown measurement angles.

2.3.4 Gooding's Method

Gooding's Assumptions

The only assumption made in Gooding's method is that the target satellite is moving under unperturbed Keplarian motion. One of Goodings's original intentions was to develop a method that was independent as possible from any details of the observer [12]. Because of this, Gooding's method has an advantage of being able to use measurements from multiple revolutions of a satellite. It also does not exhibit the singularities that Gauss' and Laplace's methods do for the co-planar and circular orbits [13]. **Gooding Method Overview**

Gooding starts by setting up the vector geometry of the AIOD problem using the observed line of sight vector ($\lambda_\mathbf{j}$), observed position ($\mathbf{R_j}$), and satellite position ($\mathbf{r_j}$), at three distinct

times, t_j ($j = 1, 2, 3$). Assuming the unknown range, ρ_j, then the observation geometry at time, t_j, is given by Equations (2.29) and (2.30).

$$\rho_j \lambda_j = \rho_j \tag{2.29}$$

$$\rho_j = r_j - R_j \tag{2.30}$$

Two initial guess must be made to initialize the algorithm. The first guess if for the edge ranges, ρ_j, for $j = 1, 3$, the second is whether the target orbit is prograde or retrograde. Gooding then solves Lambert's Problem using the initial range estiamtes to aquire an estimated position vector at t_2. Gooding used a custom Lambert solver in the development of this method, but any accurate Lambert solver can be used.

If the position vector is correct for the target orbit, then the unit vector, $\hat{\mathbf{r_2}}$, would perfectly match with the second measured LOS, λ_2. Unfortunately due to measurement error and imperfect initial guess, this almost never happens, so an iterative process must be used. Gooding used a Newton-Raphson algorithm to iteratively correct the range values until a solution converges. Following is a detailed algorithmic design of Gooding's AIOD method.

Gooding Method Pseudo-code

---

1. Givens: $\lambda_\mathbf{j}$, $\mathbf{R_j}$, t_j, for $j = 1, 2, 3$.

2. Using measured LOS vectors, $\hat{\lambda}_\mathbf{j}$, and the given information, form the main vector equation of the AIOD process.

$$\rho_j \hat{\lambda}_j = \rho_j = \mathbf{r_j} - \mathbf{R_j} \tag{2.31}$$

3. Gooding uses the following notation, $t_{jk} = t_k - t_j$, and the assumption that time t_j takes the form $t_1 < t_2 < t_3$. This gives t_{jk} a "special role" for the measurement at t_j for $j = 2$, which Gooding chose as the time to compare the iterative results with.

4. Intitalize a first guess for edge ranges, ρ_1 and ρ_3. Any known information about the observed satellites can be used to suitably bound the guess on the range values. If no information on the observed satellites is available, a reasonable initial guess is a multiple of Earth's radius.

5. Assume a direction (prograde or retrograde) for the orbit being estimated.

6. Begin iterative loop. End once tolerance for range changing between iterations is met.

   (a) Used the assumed ranges at t_1 and t_3 to form the satellite position vectors r_1 and r_3 from Equation(2.31).

   (b) Solve Lambert's Problem with r_1, r_3, t_{31} to estimate r_2.

   (c) Define target functions f and g to use in the iterative correction process. To identify the normal to a family of parallel planes, Gooding uses LOS at t_2 and $\hat{\lambda}_{\mathbf{2}}$. The origin of a unique coordinate system within each plane is the intersection of $\hat{\lambda}_{\mathbf{2}}$ with each plane.

   (d) Compute the origin (P_c) of the above mentioned plane. The target functions are then the coordinates (f, g) in the plane. For each iteration the axis system is refined and results in different values of the edge ranges (ρ_1, ρ_2) and the target functions. To simplify notation in the following analysis, Gooding makes the following changes: $\rho_1 = x$ and $\rho_2 = y$.

   (e) To correct the assumed values of ρ_1 and ρ_3, employ a Newton-Raphson protocol.

$$\begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = - \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}^{-1} \begin{pmatrix} f \\ g \end{pmatrix} \tag{2.32}$$

   (f) Gooding defines δ_x, δ_y as the corrections to the estimate of a pair of roots of the following two equations:

$$f(x, y) = 0 \tag{2.33}$$

$$g(x, y) = 0 \tag{2.34}$$

(g) Assume $g = 0$. The allows Equation (2.32) to be broken into two simpler equations, where $D = f_x g_x y - f_y g_x$ is the determinant of the derivative matrix in Equation (2.32).

$$\delta_x = -D^{-1} f g_y \tag{2.35}$$

$$\delta_y = -D^{-1} f g_x \tag{2.36}$$

(h) The partial derivatives of f and g are found by truncating a Taylor series expansion of f and g after the linear terms. This introduces some error into the process, but as long as the initial values for x and y are not too far from the solution and the determinant matrix is well conditioned, then the iteration process will display quadratic convergence [12]. If the initial guess is too far from the true solution and/or the derivative matrix is ill conditioned, then the process may require a large number of iterations to ensure convergence or the solution may not converge at all [13].

(i) The correction procedure used in this method sometimes provides multiple solutions, so Gooding introduced another Newton Raphson like procedure that guides the convergence away from the previously known solutions to obtain all available solutions. Gooding handles the multi-revolution issue of this method by changing the increment sizes upon the failure to obtain a solution to Lambert's problem after a large Newton correction to the ranges.

7. Once range values have reached the specified tolerance, the process is complete and the orbit is known by the state $\{\mathbf{r_2}, \mathbf{v_2}\}$.

2.4 Particle Swarm Optimization

Particle Swarm Optimization (PSO) is a stochastic numerical method of estimation inspired by the formation behavior often seen in birds. The PSO algorithm works by sharing information among members, or particles, of the swarm through "collective intelligence"[14][15][16] . As a population based algorithm, PSO starts by initializing a set of M particles with N dimensional position that represent potential state values. Each particle has an associated velocity in the search space, which controls the update of the particle position, and an associated cost given its current position in the N dimensional search space. The particles are weighted to balance their current "best" and being pulled towards the current global "best" optimal position. A randomized search factor is also included to each particle's update to add to the stochastic nature of the algorithm.

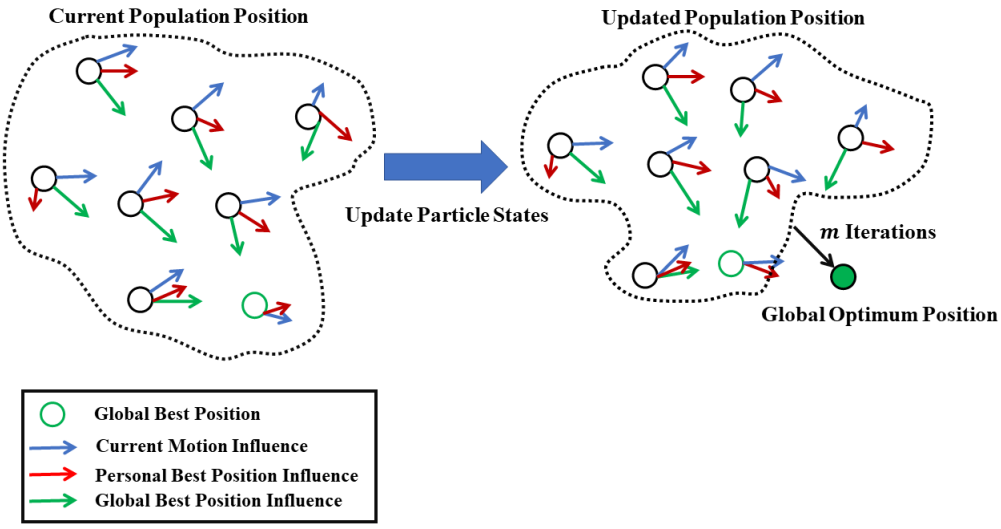


Figure 2.4 General representation of how particles move in PSO. Particle motion is influenced by the current best position (blue arrows), current motion (red arrows), and current global best position (green arrows). The population converges to the global optimal position after m iterations.

Figure 2.4 shows the general concept behind PSO and how the particles move within the system [1]. In this research 50 observation measurements were made during a simulation time equivalent to the period of the observer orbit and run through 50 iterations of the PSO.

3 Problem Definition and Method of Solving

The purpose of the research in this thesis is to prove the method of using a particle swarm optimizer as the numerical iterator for initial orbit determination of a novel observer/target pairing, Rock Around Orbits (RAOs). Once PSO estimated an initial state, it was numerically propagated using *ode45*.

3.1 Rock Around Orbits

Rock Around Orbits are based off of flower constellation theory pioneered by Mortari [17]. The term rock-around refers to how the observer spirals around the target orbit, going inside, outside, above and under, or *rocking around*, the target orbit. The theory behind RAOs covers how to design the orbital parameters of the observer, which produce relative trajectories spiraling around the assigned target orbit. Figure 3.1 shows the RAO observer and the GEO target in the inertial frame.

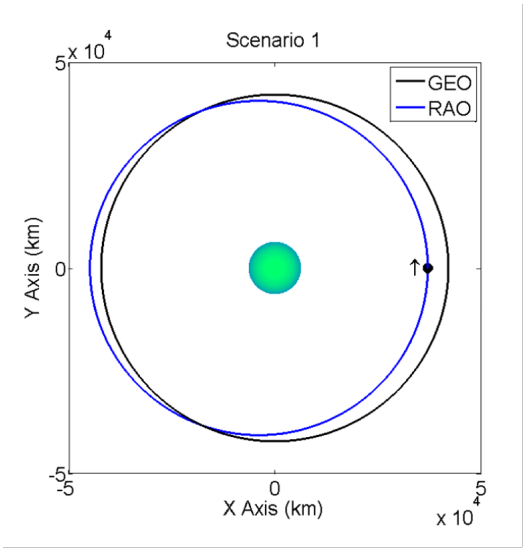


Figure 3.1 Rock Around Orbit and GEO target in the inertial frame [2].

Figures 3.2 and 3.3 show the observer and target in the target rotating frame and demonstrates the *rock-around* motion of the observer about the target orbit.

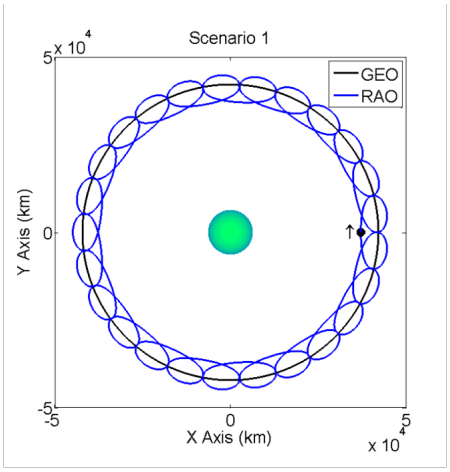


Figure 3.2 Rock Around Orbit and GEO target in the 2D target rotating frame [2].

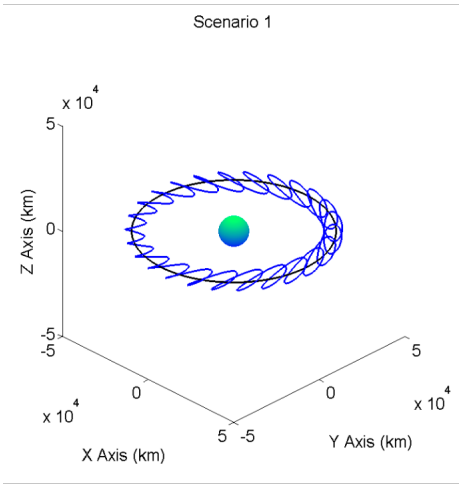


Figure 3.3 Rock Around Orbit and GEO target in the 3D target rotating frame [2].

In general the ability to design symmetric periodic solutions makes this theory a good tool for reconnaissance and space based surveillance missions planning. For a full description of the Rock Around Orbit Theory, see Bourgeois' thesis [2].

3.2 Particle Swarm Optimization

In this research two versions of the PSO algorithm was used, a two-variable and a six-variable version. Six variable PSO is based on the full state information. Two-variable PSO cost function uses the initial and final potions to estimate the velocity by solving Lambert's Problem. Algorithm 0 outlines the pseudo-code for PSO.

Algorithm 1 Particle Swarm Optimization Algorithm Pseudo-code

1: Initialize particle states: $\mathbf{x}_i$ and $\mathbf{v}_i$ for $i = 1, ..., m$
2: $\hat{\mathbf{x}}_i \leftarrow \mathbf{x}_i$ and $\hat{\mathbf{g}} = min\ J(\mathbf{x}_i)$ for $i = 1, ..., m$
3: **for** $n = 1$ to max iterations N **do**
4: **for** $i = 1$ to number particles m **do**
5: $J_i \leftarrow J(\mathbf{x}_i)$ Find current cost of particle:
6: Check if personal best cost:
7: **if** $J_i < J_{best_i}$ **then**
8: $J_{best_i} \leftarrow J_i$
9: $\mathbf{x}_{best_i} \leftarrow \mathbf{x}_i$ Update personal best state
10: **end if**
11: Randomly generate r_1, $r_2 \in U\,[0, 1]$ for particle velocity update:
12: Update particle velocity: $\mathbf{v}_i \leftarrow \omega \mathbf{v}_i + \eta_1 r_1\,(\mathbf{x}_{best_i} - \mathbf{x}_i) + \eta_2 r_2\,(\hat{\mathbf{g}} - \mathbf{x}_i)$
13: Update particle position: $\mathbf{x_i} \leftarrow \mathbf{x}_i + \mathbf{v}_i$
14: **end for**
15: Check if new global best:
16: **if** $min(J_i\,(\mathbf{x}_i)) < J\,(\hat{\mathbf{g}}_i)$ **then**
17: $\hat{\mathbf{g}}_i \leftarrow \mathbf{x}_i$
18: **end if**
19: Check if tolerance met for convergence
20: **if** $g_{error} < \tau$ (Algorithm 2) **then**
21: break
22: **end if**
23: **end for**

Lines 18-24 of Algorithm 0 were not implemented in this research, but the possibility of adding it in the future is discussed in Chapter 5.

4 Results

This chapter includes the results from the two and six variable PSO algorithms for varying amounts of noise. The algorithms both used 50 measurements over half of the period of the observer, which amounts to a measurement every 3.6° or every 864 seconds (14.4 minutes). Each subsection covers a noise level and includes the initial range and speed results propagated using two-body dynamics for the period of the observer, the initial states estimated by the PSO, the difference in the estimated and true initial states, and the percent error for estimated range and speed.

4.1 Two Variable PSO

The following figures display the results of ten runs of the two variable PSO for different noise levels. The three noise levels chosen were no noise, 5 arcseconds of noise, and 50 arcseconds of noise. Ten runs were done because there is a random component of the PSO algorithm. The dashed magenta line in each figure represents the true value. Since the true target orbit is a circular orbit, the range and speed are constant values across the entirety of the orbit. The other dashed line(s) in each figure are the run which produced the best result.

4.1.1 No Noise Results

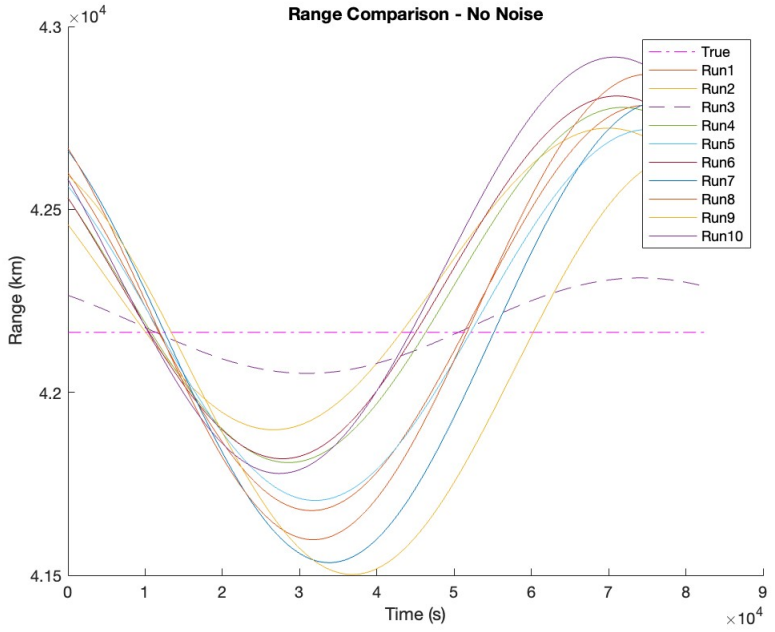


Figure 4.1 Two variable range results of ten runs. Run 3 produced the closest results to the true range

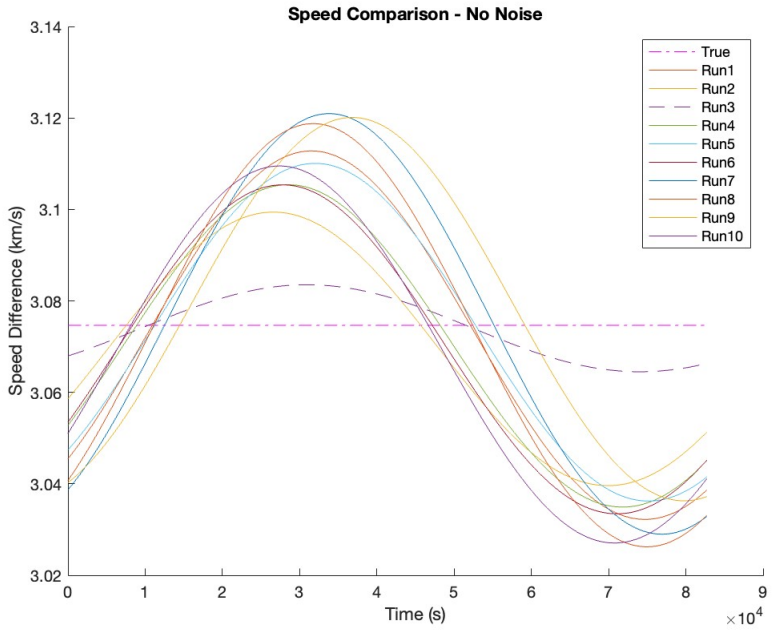


Figure 4.2 Two variable speed results of ten runs. Run 3 produced the closest results to the true speed.

Figures 4.1 and 4.2 depict that the third run produced the best results, with a percent error of approximately 0.24% for range and 0.22% for speed. Below, Table 4.1 shows the full initial state of each run, as estimated from the PSO algorithm. Table 4.2 shows the difference in the estimated initial state and the true initial state.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	42146	6206.6	-74.701	-0.4731	3.0084	-0.0265
Run2	41949	6559.7	-51.046	-0.5	3.0174	-0.0176
Run3	41672	7055.4	-17.838	-0.5194	3.0237	-0.0058
Run4	42049	6379.1	-63.141	-0.4881	3.0134	-0.0221
Run5	42095	6297.1	-68.636	-0.4769	3.0098	-0.0241
Run6	42051	6376.7	-63.302	-0.4896	3.0139	-0.0222
Run7	42228	6058.6	-84.615	-0.4593	3.0036	-0.0304
Run8	42237	6042.9	-85.665	-0.4638	3.005	-0.0309
Run9	42135	6225.7	-73.419	-0.462	3.0049	-0.0257
Run10	42121	6251	-71.722	-0.4848	3.0121	-0.0255

Table 4.1 Two Variable PSO No Noise State Results.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	622.37	1115.1	74.701	0.0608	0.0196	0.0265
Run2	425.29	762.01	51.046	0.0339	0.0106	0.0176
Run3	148.62	266.29	17.838	0.0145	0.0043	0.0058
Run4	526.05	942.57	63.141	0.0458	0.0145	0.0221
Run5	571.84	1024.6	68.636	0.057	0.0182	0.0241
Run6	527.39	944.97	63.302	0.0443	0.0141	0.0222
Run7	704.96	1263.1	84.615	0.0746	0.0243	0.0304
Run8	713.71	1278.8	85.665	0.0701	0.023	0.0309
Run9	611.68	1096	73.419	0.0719	0.0231	0.0257
Run10	597.54	1070.7	71.722	0.0491	0.0158	0.0255

Table 4.2 Two Variable PSO No Noise State Errors.

The percent errors of the initial range and speed of each run were all relatively low. The worst runs, Runs 7 and 8, had an approximate percent error of 1.18% and 1.19% for range and 1.17% and 1.10% for speed respectively. The percent errors for all ten runs can be seen in Table 4.3.

	Range % Error	Speed % Error
Run1	1.0351	0.95016
Run2	0.69854	0.52263
Run3	0.23979	0.21776
Run4	0.86959	0.7116
Run5	0.94801	0.8866
Run6	0.87189	0.68915
Run7	1.1784	1.1703
Run8	1.1937	1.1039
Run9	1.0166	1.1183
Run10	0.99222	0.76908

Table 4.3 Two Variable PSO Range and Speed Percent Errors for No Noise.

4.1.2 5 Arcseconds of Noise Results

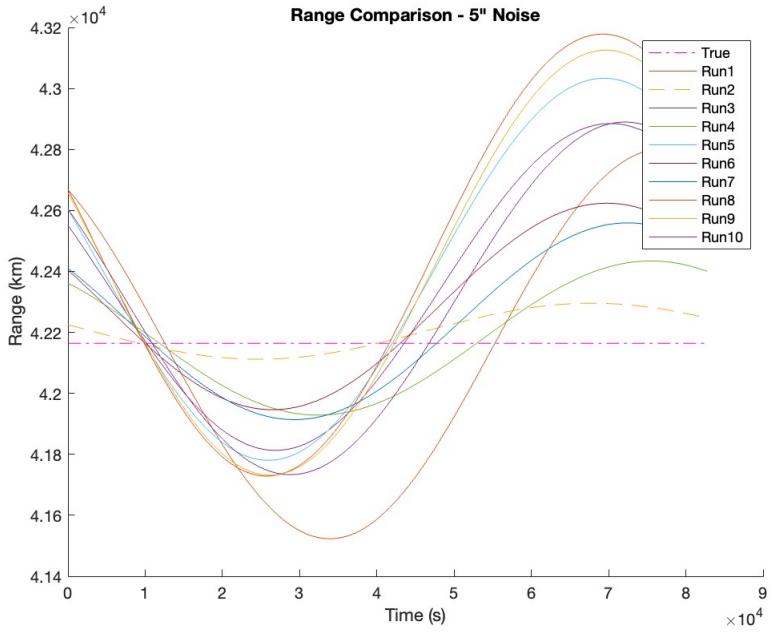


Figure 4.3 Two variable range results of ten runs. Run 2 produced the closest results to the true range

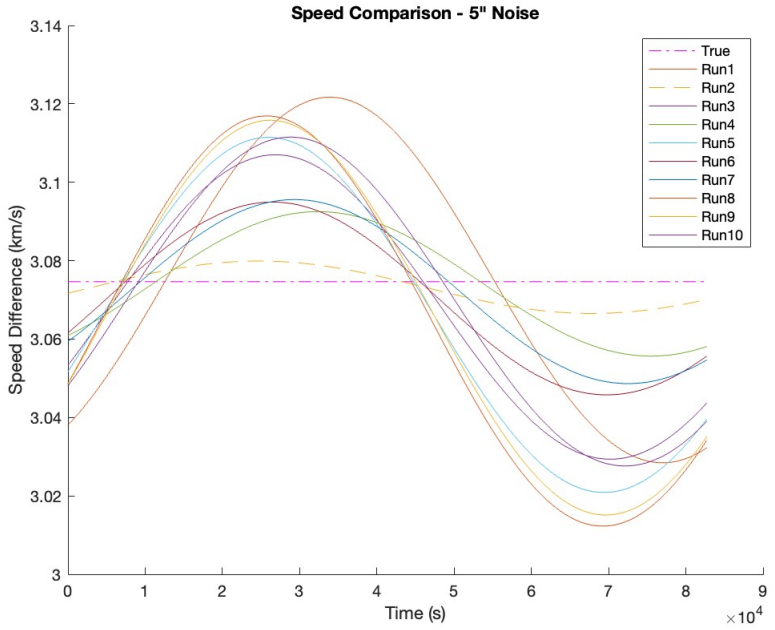


Figure 4.4 Two variable speed results of ten runs. Run 2 produced the closest results to the true speed.

Figures 4.3 and 4.4 depict that the second run produced the best results, with a percent error of approximately 0.14% for range and 0.097% for speed. Below, Table 4.4 shows the full initial state of each run, as estimated from the PSO algorithm. Table 4.5 shows the difference in the estimated initial state and the true initial state.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	42238	6041.7	-85.697	-0.4582	3.0032	-0.0308
Run2	41613	7161.4	-10.733	-0.5274	3.0261	-0.0035
Run3	42078	6327.5	-66.594	-0.4894	3.0137	-0.0235
Run4	41811	6806	-34.542	-0.5045	3.019	-0.0115
Run5	42150	6199.5	-75.184	-0.4864	3.0125	-0.0269
Run6	41873	6696.2	-41.9	-0.5061	3.0194	-0.0142
Run7	41881	6681.8	-42.865	-0.5015	3.018	-0.0145
Run8	42239	6039.1	-85.929	-0.4807	3.0104	-0.0314
Run9	42224	6066.7	-84.074	-0.4804	3.0103	-0.0306
Run10	42152	6196.4	-75.347	-0.4788	3.0102	-0.0269

Table 4.4 Two Variable PSO 5" Noise State Results.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	714.32	1280	85.697	0.0757	0.0248	0.0308
Run2	89.445	160.26	10.733	0.0065	0.0019	0.0035
Run3	554.89	994.25	66.594	0.0445	0.0142	0.0235
Run4	287.84	515.74	34.542	0.0294	0.0089	0.0115
Run5	626.33	1122.2	75.184	0.0475	0.0154	0.0269
Run6	349.09	625.49	41.9	0.0278	0.0085	0.0142
Run7	357.12	639.88	42.865	0.0324	0.01	0.0145
Run8	715.83	1282.6	85.929	0.0532	0.0176	0.0314
Run9	700.41	1255	84.074	0.0535	0.0176	0.0306
Run10	628.07	1125.3	75.347	0.0551	0.0178	0.0269

Table 4.5 Two Variable PSO 5" Noise State Errors.

The percent errors of the initial range and speed of each run were all relatively low. The worst run, Run1, had an approximate percent error of 1.19% for both range and speed. The percent errors for all ten runs can be seen in Table 4.6.

	Range % Error	Speed % Error
Run1	1.1948	1.1895
Run2	0.14376	0.096526
Run3	0.91892	0.69458
Run4	0.46863	0.44674
Run5	1.0419	0.74747
Run6	0.5706	0.42518
Run7	0.58404	0.49609
Run8	1.1974	0.84587
Run9	1.1705	0.84862
Run10	1.0449	0.86341

Table 4.6 Two Variable PSO Range and Speed Percent Errors for 5" of Noise.

4.1.3 50 Arcseconds of Noise Results

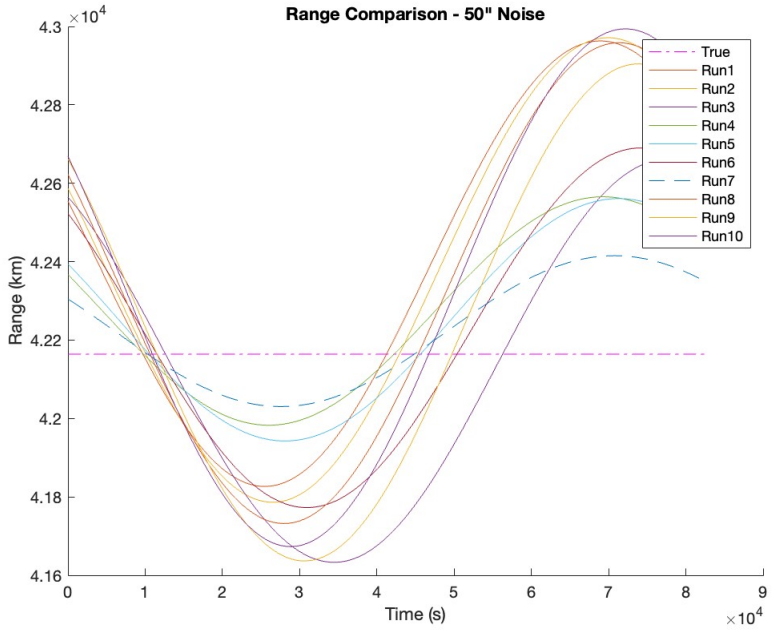


Figure 4.5 Two variable range results of ten runs. Run 7 produced the closest results to the true range

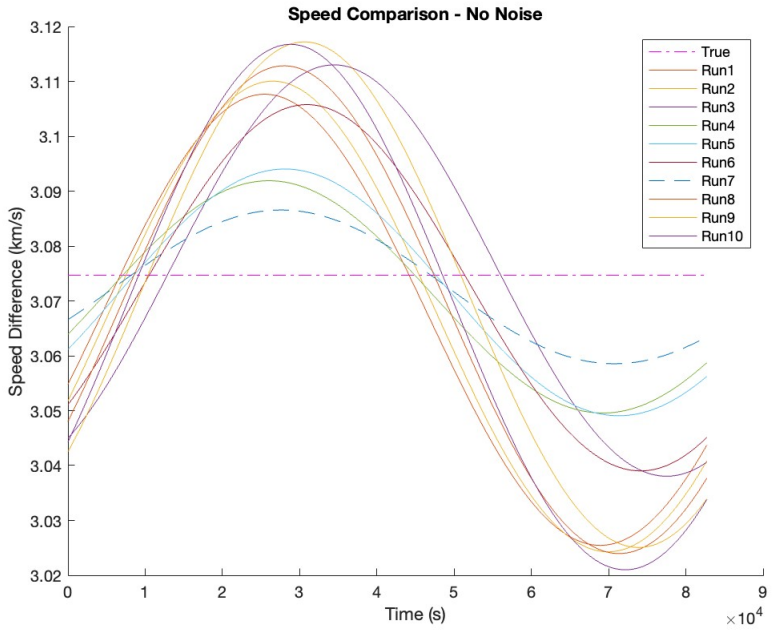


Figure 4.6 Two variable speed results of ten runs. Run 7 produced the closest results to the true speed.

Figures4.5 and 4.6 depict that the seventh run produced the best results, with a percent error of approximately 0.33% for range and 0.26% for speed. Below, Table4.7 shows the full initial state of each run, as estimated from the PSO algorithm. Table4.8 shows the difference in the estimated initial state and the true initial state.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	42082	6319.9	-67.05	-0.4925	3.0147	-0.0237
Run2	42128	6237.7	-72.696	-0.4865	3.0126	-0.0259
Run3	42096	6294.7	-68.824	-0.4717	3.0081	-0.024
Run4	41819	6791.6	-35.49	-0.5109	3.021	-0.0119
Run5	41857	6723.5	-40.093	-0.5051	3.0192	-0.0135
Run6	42039	6398.4	-61.822	-0.4844	3.0123	-0.0215
Run7	41730	6952.2	-24.761	-0.5164	3.0228	-0.0082
Run8	42177	6151.3	-78.407	-0.4788	3.01	-0.0281
Run9	42229	6057.4	-84.415	-0.4675	3.0062	-0.0305
Run10	42239	6039.7	-85.749	-0.4714	3.0074	-0.0311

Table 4.7 Two Variable PSO 50" Noise State Results.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	559.02	1001.8	67.051	0.0414	0.0132	0.0237
Run2	604.95	1084	72.696	0.0474	0.0153	0.0259
Run3	572.98	1027	68.824	0.0623	0.0198	0.024
Run4	295.81	530.06	35.49	0.023	0.007	0.0119
Run5	333.87	598.21	40.093	0.0288	0.0088	0.0135
Run6	515.31	923.26	61.822	0.0495	0.0157	0.0215
Run7	206.26	369.55	24.761	0.0175	0.0052	0.0082
Run8	653.19	1170.4	78.407	0.0551	0.0179	0.0281
Run9	706.01	1264.3	84.415	0.0664	0.0217	0.0305
Run10	715.38	1282	85.749	0.0625	0.0205	0.0311

Table 4.8 Two Variable PSO 50" Noise State Errors.

The percent errors of the initial range and speed of each run were all relatively low. The worst runs, Runs 9 and 10, had an approximate percent error of 1.18% and 1.20% for range and 1.04% and 0.99% for speed respectively. The percent errors for all ten runs can be seen in Table 4.3.

	Range % Error	Speed % Error
Run1	0.92597	0.64722
Run2	1.0049	0.74426
Run3	0.94983	0.96611
Run4	0.48185	0.34984
Run5	0.5452	0.43925
Run6	0.85128	0.76727
Run7	0.33406	0.26364
Run8	1.0884	0.86638
Run9	1.1805	1.0461
Run10	1.1966	0.98702

Table 4.9 Two Variable PSO Range and Speed Percent Errors for 50" of Noise.

4.1.4 100 Varying Noise Runs

The results in this section are from 100 runs with an incremental noise increase, adding one arcsecond of noise each run, until a noise level of 100 arcseconds was reached.The best result of the 100 runs with varying noise for range was with 24 arc-seconds of noise and for speed was with 40 arc-seconds of noise. Table 4.10 showcases the best and worst runs. Even the worst runs for both range and speed had a relatively low percent error, with the worst error being about 1.3%.

	Noise Level (Arcseconds)	Value	% Error
Best Range Result	24	42226 (km)	0.14801
Worst Range Result	10	42675 (km)	1.2123
Best Speed Result	40	3.0711 (km/s)	0.11529
Worst Speed Result	3	3.0354 (km/s)	1.2765

Table 4.10 Two Variable PSO best and worst range and speed results after 100 runs with varying noise.

4.2 Six Variable PSO

The results of the six variable PSO algorithm is in the same format of the two variable PSO algorithm above. The following figures display the results of ten runs of the six variable PSO for different noise levels. The same three noise levels at the two variable PSO, no noise, 5 arcseconds of noise, and 50 arcseconds of noise, were run. Again, ten runs were done

because of the random component of the PSO algorithm. The dashed magenta line in each figure represents the true value. Since the true target orbit is a circular orbit, the range and speed are constant values across the entirety of the orbit. The other dashed line(s) in each figure are the run which produced the best result.

4.2.1 No Noise Results

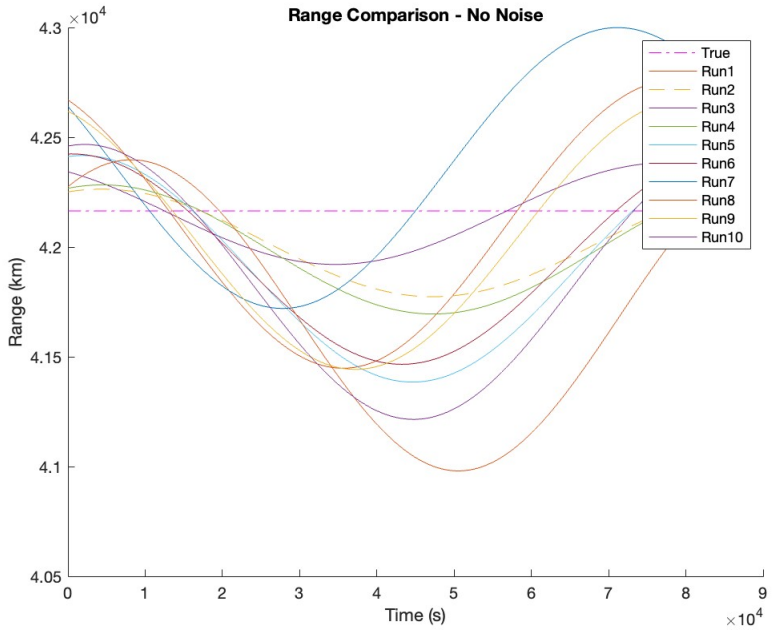


Figure 4.7 Six variable range results of ten runs. Run 3 produced the closest results to the true range

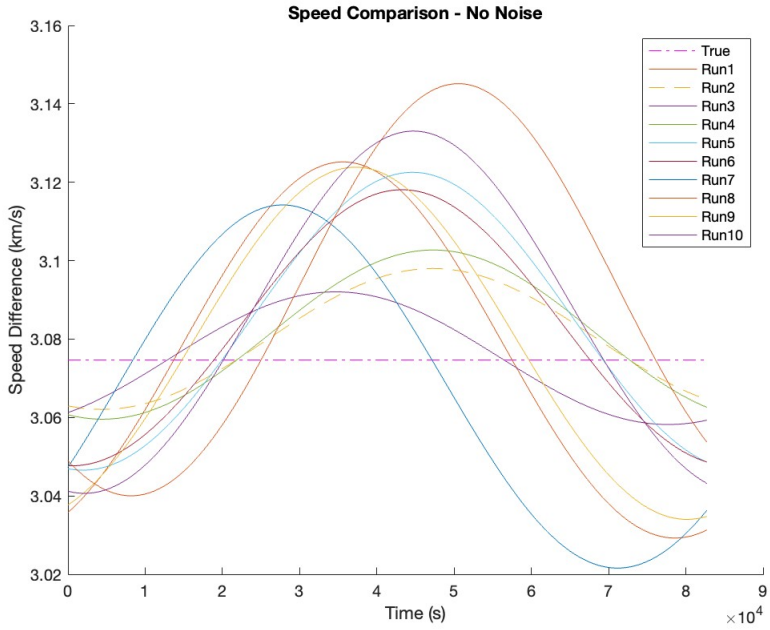


Figure 4.8 Six variable speed results of ten runs. Run 3 produced the closest results to the true speed.

Figures4.7 and 4.8 depict that the second run produced the best results, with a percent error of approximately 0.21% for range and 0.38% for speed. Below, Table4.11 shows the full initial state of each run, as estimated from the PSO algorithm. Table4.18 shows the difference in the estimated initial state and the true initial state.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	42241	6036.6	-86.088	-0.4533	3.0016	-0.0309
Run2	41652	7091.9	-15.397	-0.5086	3.0205	-0.0048
Run3	41784	6854.8	-31.277	-0.5051	3.0193	-0.0103
Run4	41675	7049.4	-18.241	-0.5037	3.019	-0.0057
Run5	41883	6678.3	-43.101	-0.4746	3.0097	-0.0141
Run6	41899	6648.2	-45.118	-0.4764	3.0102	-0.0149
Run7	42200	6108.6	-81.263	-0.4775	3.0095	-0.0293
Run8	41685	7031.5	-19.437	-0.4784	3.0112	-0.0058
Run9	42171	6161.8	-77.697	-0.4567	3.003	-0.0274
Run10	41950	6557.1	-51.222	-0.4626	3.0058	-0.017

Table 4.11 Six Variable PSO No Noise State Results.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	717.23	1285.1	86.088	0.0807	0.0264	0.0309
Run2	128.28	229.85	15.397	0.0253	0.0075	0.0048
Run3	260.58	466.91	31.277	0.0288	0.0087	0.0103
Run4	151.98	272.31	18.241	0.0302	0.009	0.0057
Run5	359.09	643.41	43.101	0.0593	0.0183	0.0141
Run6	375.9	673.53	45.118	0.0575	0.0177	0.0149
Run7	677.04	1213.1	81.263	0.0564	0.0184	0.0293
Run8	161.94	290.16	19.437	0.0556	0.0167	0.0058
Run9	647.33	1159.9	77.697	0.0772	0.0249	0.0274
Run10	426.75	764.64	51.222	0.0713	0.0222	0.017

Table 4.12 Six Variable PSO No Noise State Errors.

The percent errors of the initial range and speed of each run were all relatively low. The worst run, Run 1, had an approximate percent error of 1.2% for range and 1.3% for speed. The percent errors for all ten runs can be seen in Table4.13.

	Range % Error	Speed % Error
Run1	1.1999	1.2647
Run2	0.2067	0.38011
Run3	0.42352	0.43629
Run4	0.24526	0.4542
Run5	0.58733	0.90221
Run6	0.61549	0.87687
Run7	1.1298	0.89012
Run8	0.26151	0.83442
Run9	1.0782	1.2022
Run10	0.70102	1.088

Table 4.13 Six Variable PSO Range and Speed Percent Errors for No Noise.

4.2.2 5 Arcseconds of Noise Results

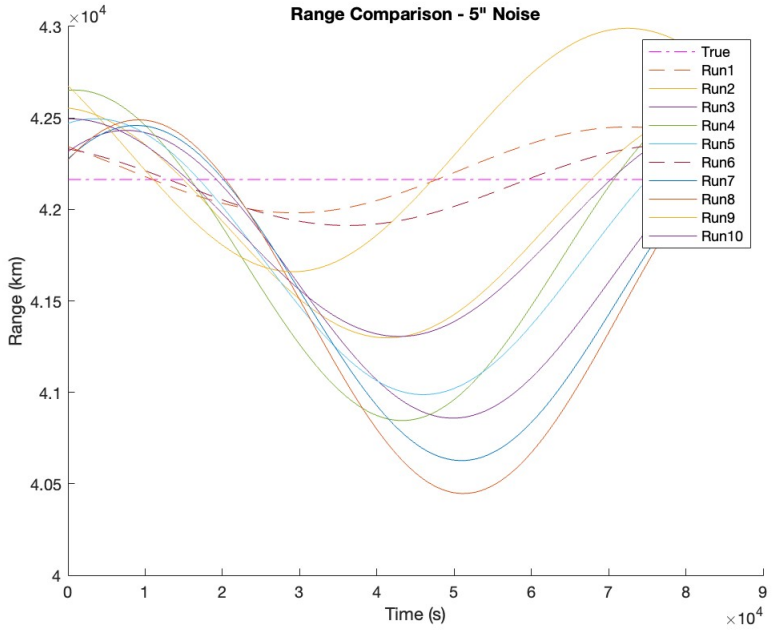


Figure 4.9 Two variable range results of ten runs. Runs 1 and 6 produced the closest results to the true range

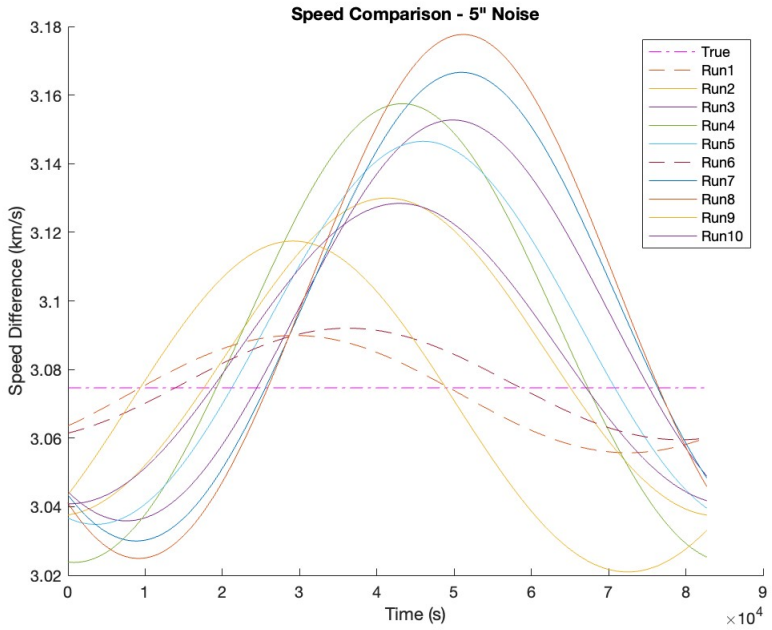


Figure 4.10 Two variable speed results of ten runs. Runs 1 and 6 produced the closest results to the true speed.

Figures4.9 and 4.10 depict that the first and sixth run produced the best results when propagated, with a percent error of approximately 0.4% and 0.39% for range and 0.36% and 0.43% for speed respectively. Below, Table4.14 shows the full initial state of each run, as estimated from the PSO algorithm. Table4.15 shows the difference in the estimated initial state and the true initial state.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	41785	6854	-31.326	-0.51	3.0208	-0.0104
Run2	42082	6320.7	-67.068	-0.4558	3.0031	-0.023
Run3	41745	6924.1	-26.627	-0.4684	3.0081	-0.0081
Run4	42215	6083.1	-82.96	-0.4275	2.9934	-0.029
Run5	41966	6529.5	-53.07	-0.453	3.0027	-0.0175
Run6	41767	6884.8	-29.275	-0.5055	3.0194	-0.0096
Run7	41687	7029	-19.61	-0.4662	3.0075	-0.0056
Run8	41678	7045.3	-18.52	-0.461	3.0059	-0.0052
Run9	42248	6022.8	-86.971	-0.4698	3.0068	-0.0316
Run10	42000	6468.3	-57.168	-0.4622	3.0055	-0.0193

Table 4.14 Two Variable PSO 5" Noise State Results.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	261.07	467.75	31.326	0.0239	0.0072	0.0104
Run2	558.64	1001	67.068	0.0781	0.0248	0.023
Run3	221.88	397.57	26.627	0.0655	0.0199	0.0081
Run4	691.29	1238.6	82.96	0.1064	0.0346	0.029
Run5	442.16	792.24	53.07	0.0809	0.0253	0.0175
Run6	243.83	436.91	29.275	0.0284	0.0085	0.0096
Run7	163.36	292.73	19.61	0.0677	0.0204	0.0056
Run8	154.3	276.45	18.52	0.0729	0.022	0.0052
Run9	724.9	1298.9	86.971	0.0642	0.0211	0.0316
Run10	476.29	853.38	57.168	0.0717	0.0225	0.0193

Table 4.15 Two Variable PSO 5" Noise State Errors.

The percent errors of the initial range and speed of each run were all relatively low. The worst run, Run 9, had an approximate percent error of 1.2% for range and 1.01% for speed. With this run there was not a singular best answer in terms of both range and speed. If a run had a low percent error on range, it had a larger speed percent error, and vice versa.

The percent errors for all ten runs can be seen in Table4.16.

	Range % Error	Speed % Error
Run1	0.42433	0.36159
Run2	0.92535	1.2059
Run3	0.35971	0.98569
Run4	1.1546	1.6519
Run5	0.72705	1.2353
Run6	0.39585	0.42985
Run7	0.26382	1.0156
Run8	0.24905	1.0932
Run9	1.2133	1.0141
Run10	0.78486	1.0996

Table 4.16 Six Variable PSO Range and Speed Percent Errors for 5" Noise.

4.2.3 50 Arcseconds of Noise Results

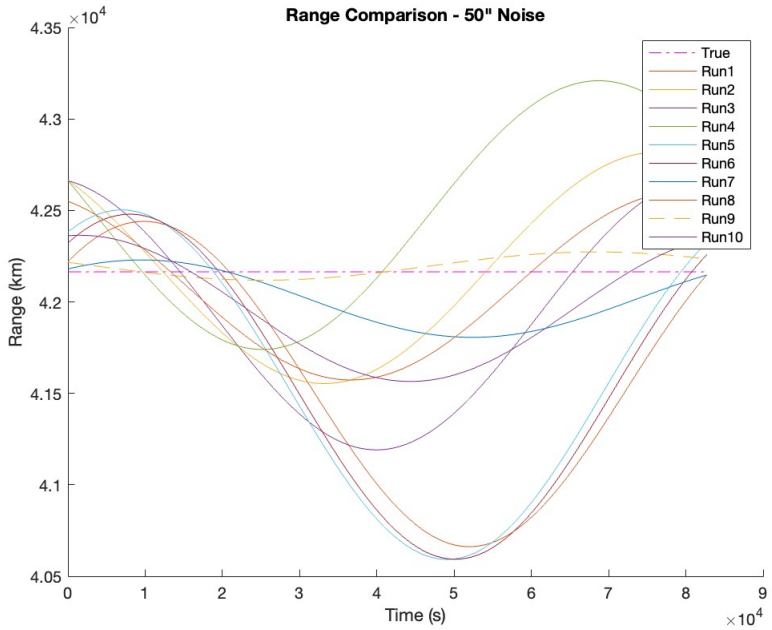


Figure 4.11 Two variable range results of ten runs. Run 9 produced the closest results to the true range

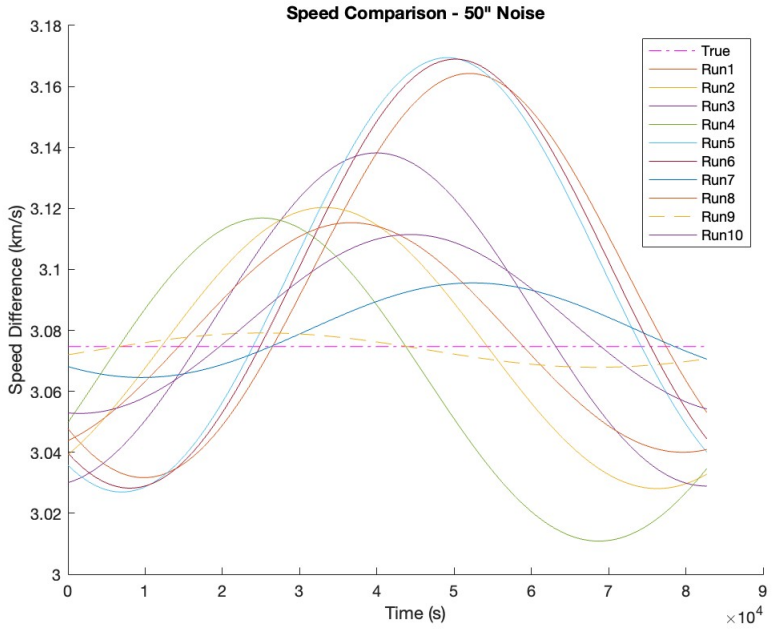


Figure 4.12 Two variable speed results of ten runs. Run 9 produced the closest results to the true speed.

Figures4.11 and 4.12 depict that the ninth run produced the best results when propagated, with a percent error of approximately 0.12% for range and 0.086% for speed. Below, Table4.17 shows the full initial state of each run, as estimated from the PSO algorithm. Table4.18 shows the difference in the estimated initial state and the true initial state.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	41606	7173.1	-9.9498	-0.4755	3.0105	-0.0025
Run2	42233	6050.1	-85.215	-0.4605	3.004	-0.0306
Run3	42230	6056.5	-84.928	-0.4409	2.9976	-0.0301
Run4	42232	6051.2	-85.321	-0.483	3.0111	-0.0311
Run5	41841	6753.5	-38.078	-0.4502	3.0022	-0.0119
Run6	41754	6908.7	-27.675	-0.4585	3.005	-0.0083
Run7	41547	7279.1	-2.8485	-0.5197	3.0238	-0.0008
Run8	42076	6330.9	-66.377	-0.4691	3.0074	-0.0231
Run9	41601	7183.2	-9.301	-0.5281	3.0263	-0.003
Run10	41811	6805.9	-34.533	-0.4873	3.0138	-0.0112

Table 4.17 Six Variable PSO 50" Noise State Results.

	X (km)	Y (km)	Z (km)	$\dot{X}$ (km/s)	$\dot{Y}$ (km/s)	$\dot{Z}$ (km/s)
Run1	82.883	148.6	9.9498	0.0584	0.0175	0.0025
Run2	709.78	1271.6	85.215	0.0734	0.024	0.0306
Run3	706.28	1265.2	84.928	0.093	0.0303	0.0301
Run4	708.96	1270.5	85.321	0.0509	0.0168	0.0311
Run5	317.1	568.18	38.078	0.0837	0.0258	0.0119
Run6	230.59	412.98	27.675	0.0754	0.023	0.0083
Run7	23.778	42.613	2.8485	0.0142	0.0042	0.0008
Run8	553.03	990.82	66.377	0.0648	0.0206	0.0231
Run9	77.306	138.51	9.301	0.0059	0.0017	0.003
Run10	287.87	515.84	34.533	0.0466	0.0142	0.0112

Table 4.18 Six Variable PSO 50" Noise State Errors.

The percent errors of the initial range and speed of each run were all relatively low. The worst run, Run 3, had an approximate percent error of 1.18% for range and 1.45% for speed. The percent errors for all ten runs can be seen in Table4.19.

	Range % Error	Speed % Error
Run1	0.13312	0.8746
Run2	1.1869	1.1536
Run3	1.1808	1.4517
Run4	1.1854	0.80935
Run5	0.51725	1.2645
Run6	0.37412	1.1347
Run7	0.038047	0.21237
Run8	0.91578	1.0028
Run9	0.12415	0.086482
Run10	0.46866	0.70649

Table 4.19 Six Variable PSO Range and Speed Percent Errors for 50" Noise.

4.2.4 100 Varying Noise Runs

The results in this section are from 100 runs with an incremental noise increase, adding one arcsecond of noise each run, until a noise level of 100 arcseconds was reached. The best result of the 100 runs with varying noise for range was with 45 arc-seconds of noise and for speed was with 29 arc-seconds of noise. Table 4.20 showcases the best and worst runs. Even the worst runs for both range and speed had a relatively low percent error, with the worst

error being about 4%.

	Noise Level(Arcseconds)	Value	% Error
Best Range Result	45	42178 (km)	0.098503
Worst Range Result	77	42668 (km)	3.4844
Best Speed Result	29	3.0708 (km/s)	0.3134
Worst Speed Result	46	3.0198 (km/s)	4.0473

Table 4.20 Six Variable PSO minimum and maximum range and speed after 100 runs with varying noise.

4.3 Varying Number of Particles and Iterations

To prove that the number of particles and iterations of the PSO algorithms used in the previous sections of Chapter 4 were sufficient, three more cases of both the two-variable and six-variable PSO algorithms were run. The first with 50 particles and 1000 iterations, the second with 1000 particles and 50 iterations, and the final being 1000 particles and 1000 iterations. In these three cases 50 arcseconds of noise were used since it provided to best results for both the two and six variable PSO runs in which 50 particles and 50 iterations were used.

Particle No. /Iteration No.	Best Range % Error	Worst Range % Error	Best Speed % Error	Worst Speed % Error
50/50	0.33406	1.1805	0.26364	1.0461
50/1000	0.57368	1.1721	0.44102	1.0164
1000/50	0.45554	1.1828	0.33527	0.92512
1000/1000	0.30567	1.0407	0.22557	1.3367

Table 4.21 Two - Variable PSO best and worst range and speed percent errors for varying particle and iteration cases.

Particle No. /Iteration No.	Best Range % Error	Worst Range % Error	Best Speed % Error	Worst Speed % Error
50/50	0.038047	1.1808	0.21237	1.4517
50/1000	0.16437	1.1798	0.38799	1.0312
1000/50	0.57656	0.96468	0.6491	0.83112
1000/1000	0.094435	1.1292	0.48648	1.7082

Table 4.22 Six - Variable PSO best and worst range and speed percent errors for varying particle and iteration cases.

4.3.1 50 Particles 1000 Iterations

This section includes both the two and six variable PSO runs with 50 particles and 50 iterations. For the two-variable PSO, the likeness of the range results can be determined by comparing Figure4.5 to Figure4.13. The same for the two-variable PSO speed results can be determined by comparing Figure 4.6 to Figure4.14.

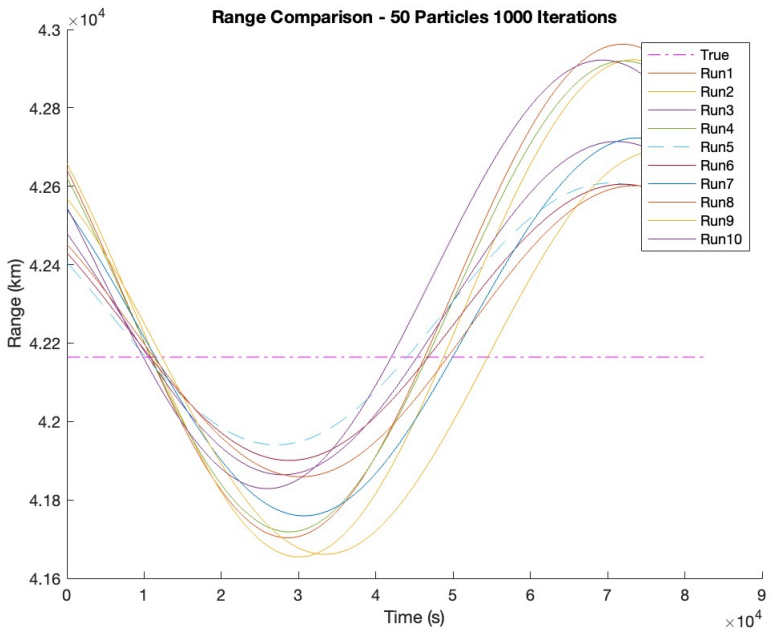


Figure 4.13 Two variable range results of ten runs with 50 particles and 1000 iterations.

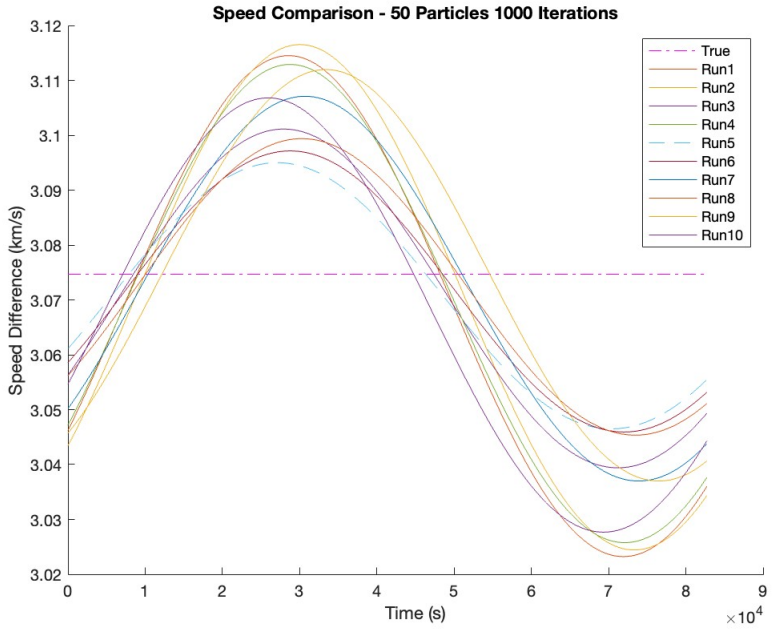


Figure 4.14 Two variable speed results of ten runs with 50 particles and 1000 iterations.

For the six-variable PSO, the likeness of the range results can be determined by comparing Figure4.11 to Figure4.15. The same for the six-variable PSO speed results can be determined by comparing Figure 4.12 to Figure4.16.

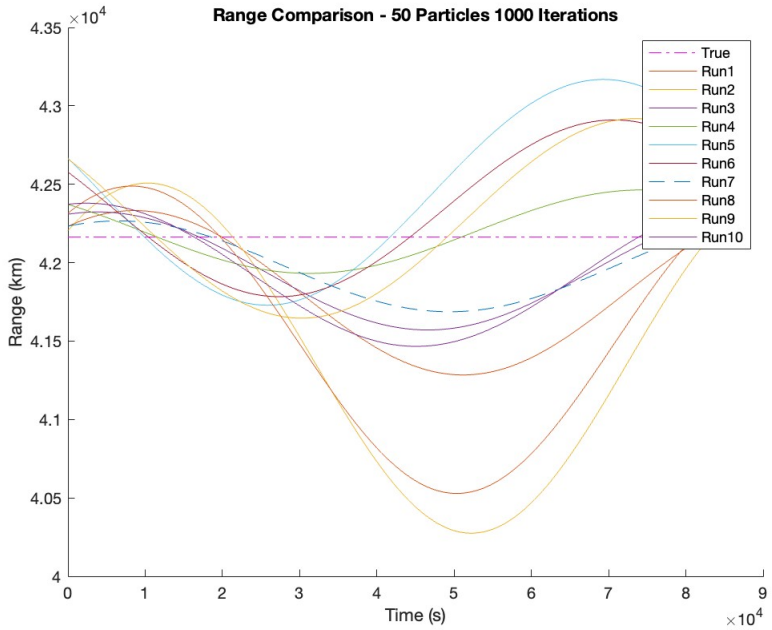


Figure 4.15 Six variable range results of ten runs with 50 particles and 1000 iterations.

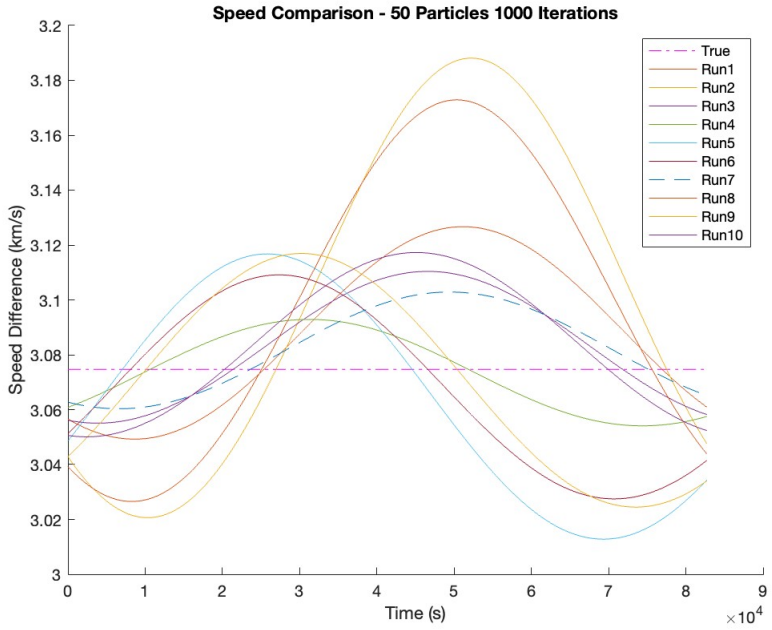


Figure 4.16 Six variable speed results of ten runs with 50 particles and 1000 iterations.

4.3.2 1000 Particles 50 Iterations

This section includes both the two and six variable PSO runs with 1000 particles and 50 iterations. For the two-variable PSO, the likeness of the range results can be determined by comparing Figure4.5 to Figure4.17. The same for the two-variable PSO speed results can be determined by comparing Figure 4.6 to Figure4.18.

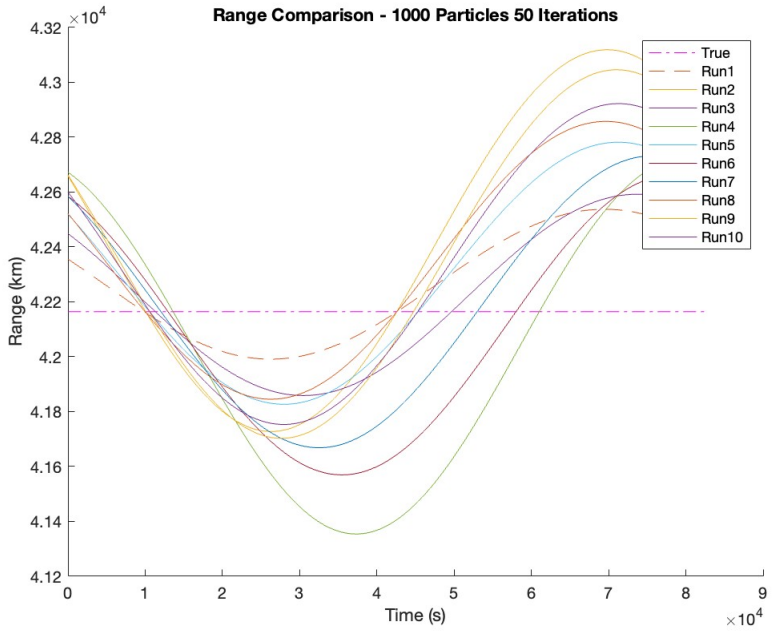


Figure 4.17 Two variable range results of ten runs with 1000 particles and 50 iterations.

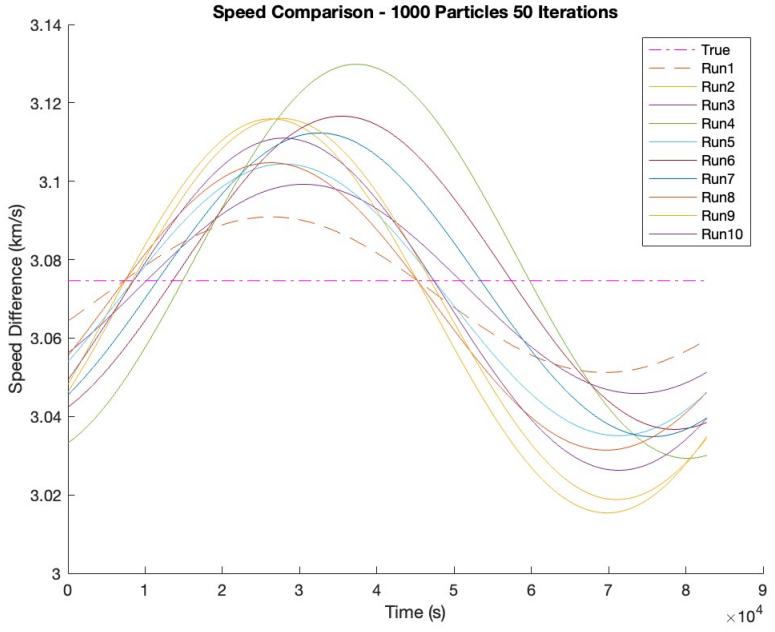


Figure 4.18 Two variable speed results of ten runs with 1000 particles and 50 iterations.

For the six-variable PSO, the likeness of the range results can be determined by comparing Figure4.11 to Figure4.19. The same for the six-variable PSO speed results can be determined

by comparing Figure 4.12 to Figure4.20.

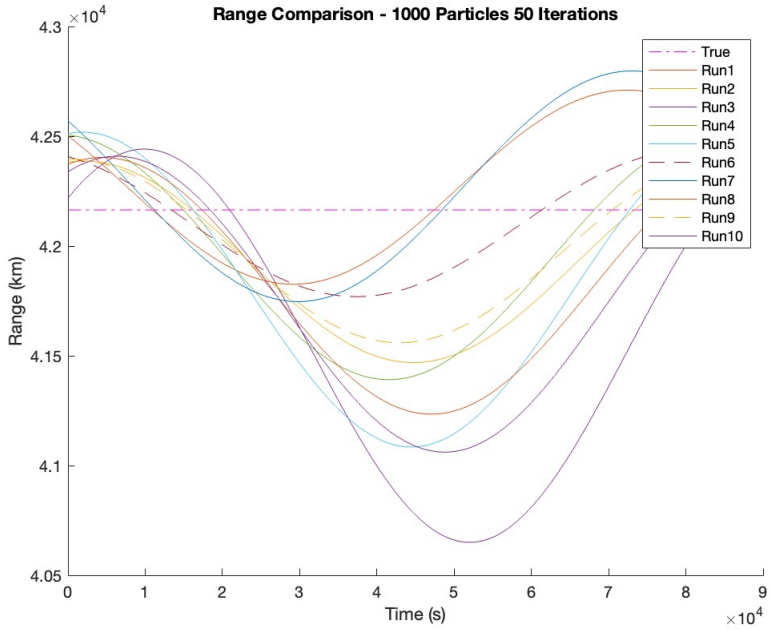


Figure 4.19 Six variable range results of ten runs with 1000 particles and 50 iterations.

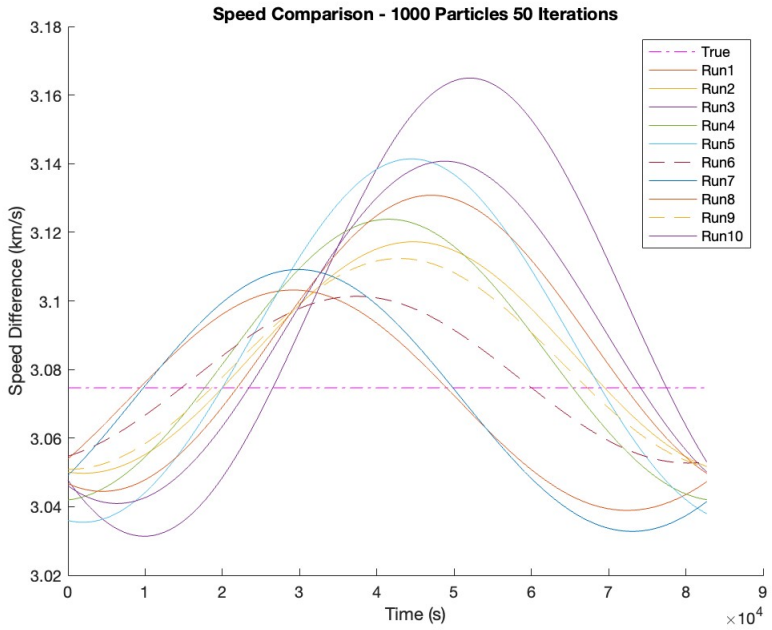


Figure 4.20 Six variable speed results of ten runs with 1000 particles and 50 iterations.

4.3.3 1000 Particles 1000 Iterations

This section includes both the two and six variable PSO runs with 1000 particles and 1000 iterations. For the two-variable PSO, the likeness of the range results can be determined by comparing Figure4.5 to Figure4.21. The same for the two-variable PSO speed results can be determined by comparing Figure 4.6 to Figure4.22.

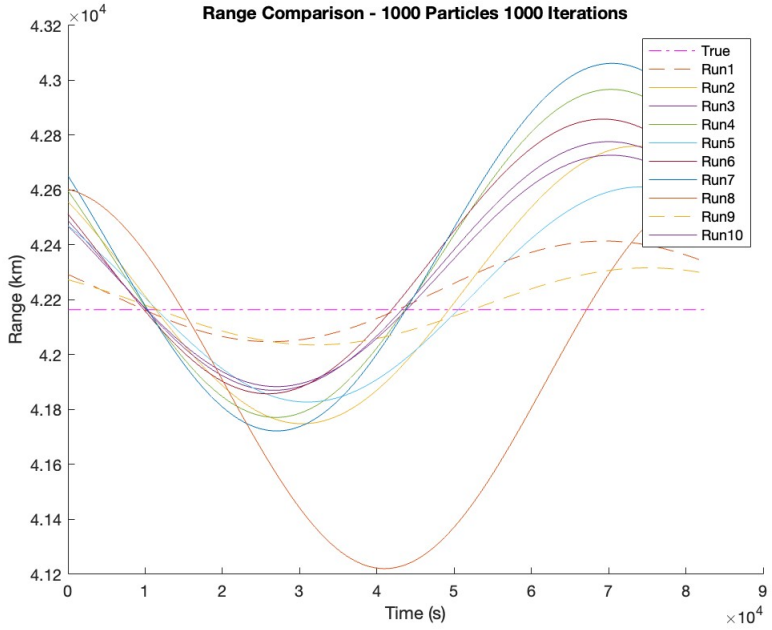


Figure 4.21 Two variable range results of ten runs with 1000 particles and 1000 iterations.

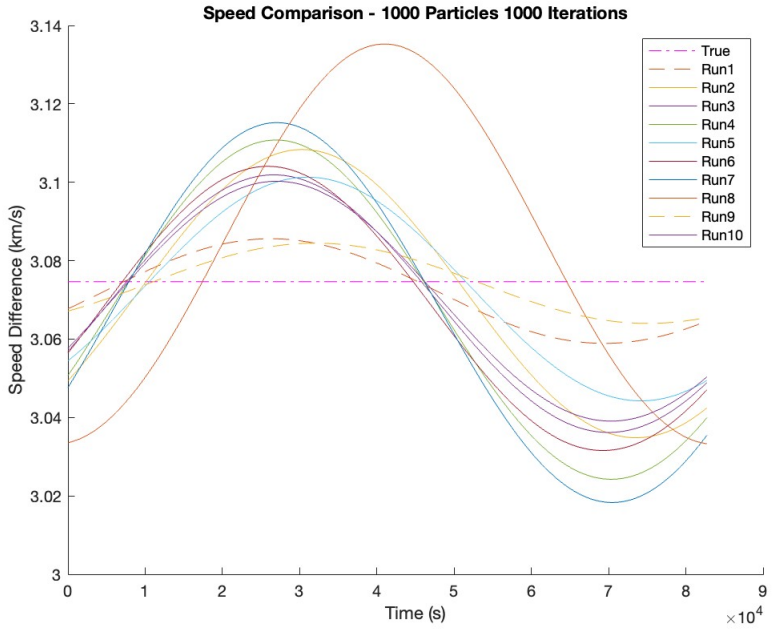


Figure 4.22 Two variable speed results of ten runs with 1000 particles and 1000 iterations.

For the six-variable PSO, the likeness of the range results can be determined by comparing Figure4.11 to Figure4.23. The same for the six-variable PSO speed results can be determined by comparing Figure 4.12 to Figure4.24.

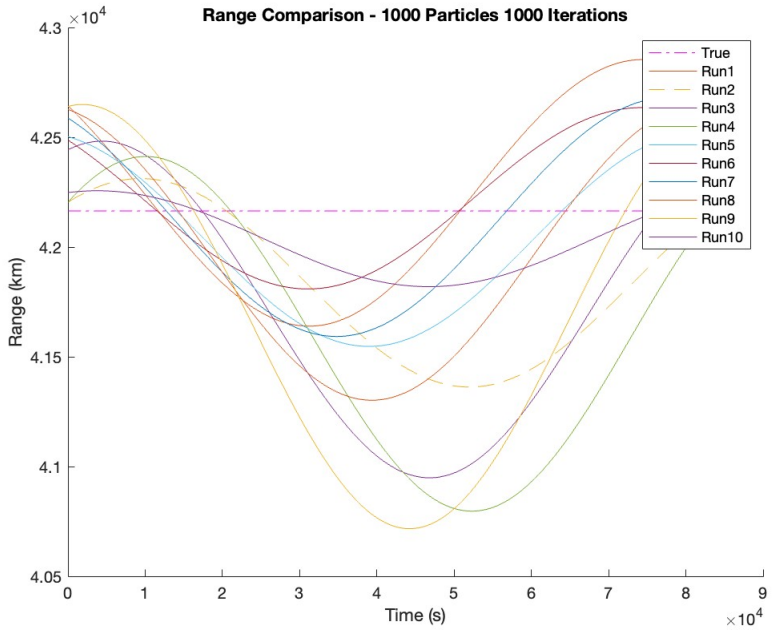


Figure 4.23 Six variable range results of ten runs with 1000 particles and 50 iterations.

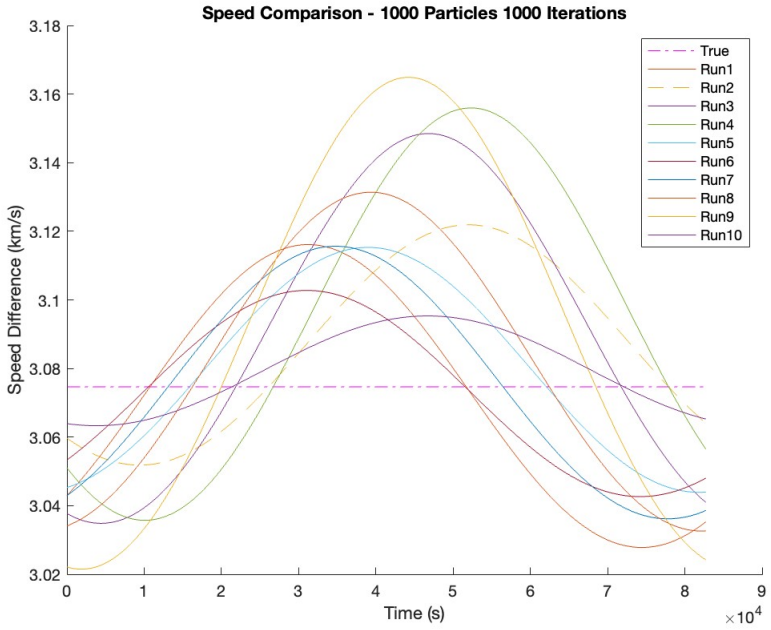


Figure 4.24 Six variable speed results of ten runs with 1000 particles and 50 iterations.

5 Conclusion & Future Work

In all noise cases for both the two and six variable versions of PSO, initial states were produced that were within 2% of the truth. From this low percent error, one can conclude that both PSO algorithms can successfully estimate an initial state of the target orbit from observations of an rock around orbit.

5.1 Comparison of Two and Six Variable PSO Results

The best and worst cases of the 100 runs for both the two variable and six variable PSO show that the best cases were about the same across both algorithms, but the worst cases for the six variable PSO had about 2.5% more error than the worst cases for the two-variable PSO. Given the sequentially increasing amount of noise of each run, that fact that the best and worst results for both range and speed were not the runs that had the lowest and highest amount of noise demonstrates the stochastic nature of using PSO as an estimator. One conclusion drawn from the oscillatory nature of the speed and range results is that the rock around orbit introduces an observation bias. The target orbit is a circular orbit with no inclination, but the observer oscillates about the region of interest that the target resides in. This oscillation around the target and slight inclination of the observer's orbit leads to the observation bias.

5.2 Future Work

There are a few additions to this method that could be added in the future to improve the accuracy and robustness. To combat the observation bias caused by the oscillation of the observer, adding a filter, such as a Kalman filter, could decrease the oscillation of the result. For future implementation, a tolerance limit could be added to the PSO algorithm. This would result in the estimated initial states to be within a specified region of the known truth and would eliminate the outliers.

The current research published on RAOs only covers two-body dynamics, so in the future I would like to extend this work to three-body dynamics, specifically the cislunar space. With more and more missions being planned to go to the Moon by both government and

commercial space agencies, monitoring the cislunar space is becoming vital to ensure the safety of the manned and unmanned mission. Projects like AFRL's ORCALE, which plans to put a satellite in orbit about an Earth-Moon Lagrange Point to monitor both natural and man-made satellites headed to the Moon, are in development to ensure the safety of the cislunar space and this method of having the observer in a RAO could increase the observation time and area. This method could easily be implemented in a number of ways and made computationally efficient for implementation onto onboard computers.

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